Projection Learning and Conservative Projection Memory

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Abstract

We introduce Projection Learning, a variational framework for runtime learning in which state updates are minimal consistency repairs rather than test-time gradient steps. Given a current state and new evidence, the update is the Bregman projection onto a consistency set, or its soft proximal analogue. We prove that this operation is characterized by a Pythagorean principle: it is exactly the update that satisfies the evidence while not increasing distance to any feasible ideal state. We instantiate the framework as Conservative Projection Memory, a neural associative state whose write direction is the innovation of a key relative to the span of previously protected evidence. This yields an exact algebraic decomposition of evidence into redundant, novel, and contradictory cases. We prove feasibility and capacity laws for exact binding, conflict impossibility, hard and soft conservative projection updates, residual contraction, and ridge and Bayesian extensions. The resulting mechanism performs runtime adaptation without test-time backpropagation while making stability, plasticity, and conflict explicit mathematical objects.

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1 Introduction

Backpropagation made hidden representations trainable by turning learning into differentiation through a static computation graph. This is the right primitive when the object being optimized is a parameterized function and the task is to distribute a scalar error through differentiable structure. Runtime learning asks for a different primitive. A deployed system receives evidence after it has already formed state, and the update must incorporate that evidence without changing what the evidence does not touch.

The central problem is therefore not only optimization but controlled repair. If the current state is s_t and the evidence at time t determines a set of acceptable states $\mathcal{C}_t$, the natural update is

$$s_{t+1} = \operatorname*{argmin}_{s \in \mathcal{C}_t} D_\psi(s, s_t).$$

If the evidence is noisy, the corresponding soft update is

$$s_{t+1} = \operatorname*{argmin}_{s} \{D_\psi(s, s_t) + \eta_t \ell_t(s)\}.$$

This paper studies these updates as a runtime learning principle. We call the principle Projection Learning: learning is minimal consistency repair.

The first contribution is a characterization theorem. Bregman projection is not merely one possible update. It is exactly the update that satisfies the evidence and obeys a Pythagorean inequality against every feasible ideal state. Thus the variational rule is equivalent to a stability statement: after projection, the learner has not moved away from any state that also satisfies the evidence.

The second contribution is a canonical neural realization called Conservative Projection Memory. A runtime memory is a matrix $W_t \in \mathbb{R}^{m \times r}$, and evidence is a key-value constraint $W k_t = v_t$. If old keys form K_{t-1}, the correct write direction is not k_t but the innovation

$$z_t = (I - \Pi_{t-1}) k_t, \qquad \Pi_{t-1} = K_{t-1} K_{t-1}^{+}.$$

The soft update is

$$W_{t+1} = W_t + \frac{\eta_t}{1 + \eta_t \|z_t\|^2} (v_t - W_t k_t) z_t^\top.$$

This is the smallest update that repairs the new residual while preserving old constraints whenever such preservation is possible.

The innovation view gives the mechanism its semantics. If $\|z_t\|$ is large, the key contains new information and the state can be updated in a new direction. If $z_t = 0$ and the implied value agrees with the evidence, the observation is redundant. If $z_t = 0$ and the value disagrees, the evidence is contradictory: no update can satisfy it while preserving the old bindings. The method therefore exposes novelty and conflict as algebraic facts, not as hidden consequences of an optimizer.

Organization. Section 2 fixes notation and the projection geometry. Section 3 states the main informal results. Section 4 gives proof sketches. Section 5 concludes. Appendix A gives a guide to the appendix; Appendix B states limitations; Appendix C discusses related work; the remaining appendices contain formal statements and proofs.

2 Preliminaries

Let S be a closed convex subset of a finite-dimensional real vector space. Let $\psi : S \to \mathbb{R}$ be differentiable and strictly convex. The Bregman divergence generated by ψ is

$$D_\psi(u, v) = \psi(u) - \psi(v) - \langle \nabla\psi(v), u - v \rangle .$$

When $\psi(s) = \|s\|_2^2/2$, this is Euclidean squared distance. All projection statements below are finite-dimensional and use the standard variational characterization of convex minimizers.

For a matrix K, K^+ denotes the Moore-Penrose pseudoinverse, $\ker K$ its nullspace, and KK^+ the orthogonal projector onto the column span of K. A memory state is a matrix $W \in \mathbb{R}^{m\times r}$. A collection of key-value evidence is written as

$$K = [k_1, \ldots, k_T] \in \mathbb{R}^{r\times T}, \qquad V = [v_1, \ldots, v_T] \in \mathbb{R}^{m\times T}.$$

Exact binding means $WK = V$. A new key k is decomposed relative to the old key span by

$$\Pi = KK^+, \qquad z = (I - \Pi)k, \qquad a = K^+k. \tag{1}$$

The vector z is the innovation. It is the component of k not explained by previous keys.

The paper uses two levels of theorem statements. Main-text theorems are informal but mathematically precise enough to communicate the result and proof method. Each such theorem points to a formal appendix theorem. The appendix theorem links back to its main-text version.

3 Main Results

3.1 Projection as the canonical repair update

Theorem 3.1 (Projection characterization, informal version of Theorem D.2)**.** *Let $\mathcal{C} \subseteq S$ be closed and convex. A point $p \in \mathcal{C}$ is the Bregman projection of s_0 onto $\mathcal{C}$ if and only if, for every $u \in \mathcal{C}$,*

$$D_\psi(u, p) + D_\psi(p, s_0) \le D_\psi(u, s_0).$$

This says that minimal repair and global stability with respect to all feasible ideal states are the same condition.

3.2 Exact binding and capacity

Theorem 3.2 (Exact associative binding, informal version of Theorem E.1)**.** *For key-value matrices $K \in \mathbb{R}^{r\times T}$ and $V \in \mathbb{R}^{m\times T}$, the constraint set $\{W : WK = V\}$ is nonempty if and only if $\ker K \subseteq \ker V$. When it is nonempty, the Euclidean projection of W_0 onto this set is*

$$W^\star = W_0 + (V - W_0K)K^+.$$

In particular, if K has full column rank, then arbitrary values can be exactly bound.

Thus an r-dimensional key space exactly stores arbitrary values for at most r independent keys. Beyond that rank, values must obey the same linear relations as keys.

3.3 Innovation, conflict, and conservative writing

Theorem 3.3 (Innovation decomposition, informal version of Theorem E.2)**.** *Assume old evidence is $WK = V$ and decompose a new key as in (1). Every old-feasible memory satisfies*

$$Wk = Va + Wz.$$

If $z = 0$, the value at k is already fixed by old evidence. If $z \neq 0$, the new evidence constrains only the innovation direction through $Wz = v - Va$.

Theorem 3.4 (Conflict impossibility, informal version of Theorem E.3)**.** *If $z = 0$, then no update preserving $WK = V$ can change Wk. Hence evidence $Wk = v$ is compatible with old evidence if and only if $v = W_t k$.*

Theorem 3.5 (Soft Conservative Projection Memory, informal version of Theorem E.5)**.** *Assume $W_t K = V$ and $z \neq 0$. The constrained soft repair problem*

$$W_{t+1} = \operatorname*{argmin}_{W:WK=V} \left\{ \frac{1}{2} \|W - W_t\|_F^2 + \frac{\eta}{2} \|Wk - v\|^2 \right\}$$

has the closed form

$$W_{t+1} = W_t + \frac{\eta}{1 + \eta \|z\|^2} (v - W_t k) z^\top.$$

It preserves old constraints and contracts the new residual by the factor $(1 + \eta \|z\|^2)^{-1}$.

3.4 Ridge and Bayesian memory

Theorem 3.6 (Ridge and Bayesian projection memory, informal version of Theorem F.3)**.** *The ridge objective*

$$J_t(W) = \frac{\lambda}{2} \|W - W_0\|_F^2 + \frac{1}{2} \sum_{\tau=1}^{t} \rho_\tau \|Wk_\tau - v_\tau\|^2$$

has minimizer $W_{t+1} = B_t A_t^{-1}$, where

$$A_t = \lambda I + \sum_{\tau=1}^{t} \rho_\tau k_\tau k_\tau^\top, \qquad B_t = \lambda W_0 + \sum_{\tau=1}^{t} \rho_\tau v_\tau k_\tau^\top.$$

The same matrix is the posterior mean and MAP estimate under a Gaussian linear observation model.

4 Technical Overview

This section sketches the proof ideas. Full formal statements and proofs appear in Appendices D, E, and F.

Theorem 4.1 (Projection characterization, proof sketch for Theorem D.2)**.** *The informal statement is Theorem 3.1.*

Proof sketch. The optimality condition for $p = \operatorname{argmin}_{s \in \mathcal{C}} D_\psi(s, s_0)$ is

$$\langle \nabla\psi(p) - \nabla\psi(s_0), u - p \rangle \geq 0 \qquad \forall u \in \mathcal{C}.$$

The Bregman three-point identity gives

$$D_\psi(u, s_0) - D_\psi(u, p) - D_\psi(p, s_0) = \langle \nabla\psi(s_0) - \nabla\psi(p), p - u \rangle .$$

The two displays are equivalent. This proves both directions. □

Theorem 4.2 (Exact binding, proof sketch for Theorem E.1)**.** *The informal statement is Theorem 3.2.*

Proof sketch. If $WK = V$ and $Ka = 0$, then

$$Va = WKa = 0.$$

Thus $\ker K \subseteq \ker V$. Conversely this nullspace condition implies

$$V = VK^+K,$$

so $W = VK^+$ is feasible. For projection, write $W = W_0 + \Delta$ and $R = V - W_0K$. The problem becomes

$$\min_\Delta \frac{1}{2} \|\Delta\|_F^2 \quad \text{subject to} \quad \Delta K = R.$$

The minimum-norm feasible solution is $\Delta = RK^+$. □

Theorem 4.3 (Innovation and conflict, proof sketch for Theorems E.2 and E.3)**.** *The informal statements are Theorems 3.3 and 3.4.*

Proof sketch. Using (1),

$$k = KK^+k + (I - KK^+)k = Ka + z.$$

If $WK = V$, then

$$Wk = WKa + Wz = Va + Wz.$$

If $z = 0$, then $Wk = Va$ for every old-feasible W. If an update Δ preserves old evidence, then $\Delta K = 0$ and hence $\Delta k = 0$ whenever $k \in \operatorname{span}(K)$. Thus contradiction cannot be repaired without breaking old evidence. □

Theorem 4.4 (Soft CPM, proof sketch for Theorem E.5)**.** *The informal statement is Theorem 3.5.*

Proof sketch. Set $W = W_t + \Delta$ and $e = v - W_tk$. Since $W_tK = V$, preserving old evidence is $\Delta K = 0$. Because $\Delta K = 0$,

$$\Delta k = \Delta(I - \Pi)k = \Delta z.$$

The objective becomes

$$\min_{\Delta : \Delta K = 0} \frac{1}{2} \|\Delta\|_F^2 + \frac{\eta}{2} \|\Delta z - e\|^2 .$$

The stationarity equation inside this subspace is

$$\Delta + \eta(\Delta z - e)z^\top = 0.$$

Solving with $\Delta = \alpha e z^\top$ yields

$$\alpha = \frac{\eta}{1 + \eta \|z\|^2}.$$

The residual satisfies

$$v - W_{t+1}k = e - \Delta z = \frac{1}{1 + \eta \|z\|^2} e.$$

□

Theorem 4.5 (Ridge and Bayesian memory, proof sketch for Theorem F.3)**.** *The informal statement is Theorem 3.6.*

Proof sketch. Differentiating J_t gives

$$0 = \lambda(W - W_0) + \sum_{\tau=1}^{t} \rho_\tau (W k_\tau - v_\tau) k_\tau^\top.$$

Rearranging,

$$W \left(\lambda I + \sum_{\tau=1}^{t} \rho_\tau k_\tau k_\tau^\top \right) = \lambda W_0 + \sum_{\tau=1}^{t} \rho_\tau v_\tau k_\tau^\top.$$

Thus $W = B_t A_t^{-1}$. Under the Gaussian prior and observation model, the negative log posterior equals $J_t(W)$ up to constants, so the posterior mean and MAP coincide with this minimizer. □

5 Conclusion

Projection Learning treats runtime learning as minimal consistency repair. Its mathematical primitive is not a gradient through a static graph but a projection of state under evidence. Conservative Projection Memory realizes this principle in a neural associative state by writing along the innovation of a key relative to protected evidence. This produces exact feasibility conditions, capacity laws, conflict detection, and closed-form stable updates. The main conceptual conclusion is that runtime learning should expose the geometry of evidence: what is already implied, what is genuinely new, and what is inconsistent with what the system has committed to preserve.

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Appendix

A Appendix Roadmap

The appendix is organized to mirror the logic of the paper while giving full formal details.

- **Part 1.** Appendix B states the limitations of the framework and clarifies what is not proved.
- **Part 2.** Appendix C gives related work after the technical story, so the main text can stay focused on definitions and results.
- **Part 3.** Appendix D proves the projection-learning facts: the Bregman identity, the projection characterization theorem, soft projection bounds, KL projection, and the gradient-descent relation.
- **Part 4.** Appendix E proves the exact memory theory: feasibility, capacity, innovation decomposition, conflict impossibility, and hard and soft conservative updates.
- **Part 5.** Appendix F proves the ridge-recursive and Bayesian forms of projection memory.

B Limitations

Projection Learning is a formal update principle, not a guarantee that every source of evidence is useful. If the value map Vh_t produces a poor target, projection will repair the wrong constraint. The theory also separates feasibility from generalization: exact binding says when constraints can be satisfied, but it does not imply that the resulting memory improves downstream prediction.

Conservative Projection Memory assumes that preserving old constraints is desirable. This is appropriate for stable binding, but nonstationary environments may require forgetting, reset, branching, or discounted ridge objectives. Exact innovation maintenance also has a computational cost. In large systems, one may use rank caps, approximate orthogonalization, diagonal or low-rank covariance estimates, or learned gates; these approximations are not covered by the exact preservation theorems.

The paper proves deterministic algebraic and convex-analytic statements. It does not prove end-to-end performance improvements for language modeling, classification under shift, or long-context retrieval. Those claims require experiments. The proposed diagnostics are residual, innovation norm, conflict rate, and contraction ratio; the theory predicts their behavior under the stated model, not their empirical distribution in trained networks.

C Related Work

Backpropagation and runtime updates. Backpropagation supplies differentiable credit assignment for static computation graphs [RHW86]. Synthetic gradients decouple modules by predicting downstream gradient information [JCO+17]. Projection Learning decouples runtime updates differently: the update is not a predicted derivative, but the solution of a local state-repair problem.

Bregman projection and online optimization. Bregman projections originate in convex feasibility and optimization [Bre67]. Online convex optimization and mirror descent use related geometries to bound regret and movement [Zin03, SS12, Haz16]. The present paper uses this geometry as a runtime-learning semantics: evidence defines consistency, and projection defines the least disruptive repair.

Passive-aggressive learning and least squares. Passive-aggressive algorithms solve minimal-change constrained updates for supervised online learning [CDK+06]. Recursive least squares and online ridge regression yield classical closed-form updates and loss bounds [Vov97, AW01]. Conservative Projection Memory adapts these ideas to internal neural state and adds the innovation/conflict decomposition induced by old evidence.

Test-time training and expressive hidden states. Test-time training updates model parameters on self-supervised objectives at inference time [SWL+20]. TTT layers make hidden states themselves trainable models updated on test sequences [SLD+25]. Conservative Projection Memory targets the same need for runtime adaptation, but replaces test-time backpropagation with closed-form projection updates.

Fast weights and linear attention. Fast weights store temporary information in rapidly changing associative matrices [BHM+16]. Linear attention can be interpreted through recurrent fast-weight updates [KVPF20, SIS21]. Conservative Projection Memory differs by writing residuals rather than raw values and by writing in the key innovation subspace rather than along the raw key.

D Projection Learning Proofs

Lemma D.1 (Bregman three-point identity)**.** *For all $a, b, u \in S$,*

$$D_\psi(u, a) - D_\psi(u, b) - D_\psi(b, a) = \langle \nabla\psi(a) - \nabla\psi(b), b - u \rangle .$$

Proof. Expand each term:

$$\begin{aligned} D_\psi(u, a) &= \psi(u) - \psi(a) - \langle \nabla\psi(a), u - a \rangle , \\ D_\psi(u, b) &= \psi(u) - \psi(b) - \langle \nabla\psi(b), u - b \rangle , \\ D_\psi(b, a) &= \psi(b) - \psi(a) - \langle \nabla\psi(a), b - a \rangle . \end{aligned}$$

Then

$$\begin{aligned} D_\psi(u, a) - D_\psi(u, b) - D_\psi(b, a) &= -\langle \nabla\psi(a), u - a \rangle + \langle \nabla\psi(b), u - b \rangle + \langle \nabla\psi(a), b - a \rangle \\ &= \langle \nabla\psi(a), b - u \rangle - \langle \nabla\psi(b), b - u \rangle \\ &= \langle \nabla\psi(a) - \nabla\psi(b), b - u \rangle . \end{aligned}$$

□

Theorem D.2 (Projection characterization, formal version of Theorem 3.1)**.** *Let S be closed and convex, let ψ be differentiable and strictly convex, let $\mathcal{C} \subseteq S$ be closed and convex, and let $s_0 \in S$. For $p \in \mathcal{C}$, the following are equivalent:*

- **Part 1.** $p = \operatorname{argmin}_{s \in \mathcal{C}} D_\psi(s, s_0)$.

- **Part 2.** *For all $u \in \mathcal{C}$,*

$$D_\psi(u, p) + D_\psi(p, s_0) \leq D_\psi(u, s_0).$$

Proof. Assume Part 1. The first-order optimality condition is

$$\langle \nabla\psi(p) - \nabla\psi(s_0), u - p \rangle \geq 0 \qquad \forall u \in \mathcal{C}.$$

Equivalently,

$$\langle \nabla\psi(s_0) - \nabla\psi(p), p - u \rangle \geq 0 \qquad \forall u \in \mathcal{C}.$$

By Lemma D.1,

$$D_\psi(u, s_0) - D_\psi(u, p) - D_\psi(p, s_0) = \langle \nabla\psi(s_0) - \nabla\psi(p), p - u \rangle .$$

Thus Part 2 follows.

Assume Part 2. The same identity implies

$$\langle \nabla\psi(s_0) - \nabla\psi(p), p - u \rangle \geq 0 \qquad \forall u \in \mathcal{C}.$$

Hence

$$\langle \nabla\psi(p) - \nabla\psi(s_0), u - p \rangle \geq 0 \qquad \forall u \in \mathcal{C}.$$

This is the variational inequality for minimizing $D_\psi(s, s_0)$ over $\mathcal{C}$. Strict convexity gives uniqueness, so Part 1 follows. □

Corollary D.3 (Fejer monotonicity and bounded drift)**.** *Let $s_{t+1} = \operatorname{argmin}_{s \in \mathcal{C}_t} D_\psi(s, s_t)$. If $u^\star \in \cap_{t=1}^T \mathcal{C}_t$, then*

$$D_\psi(u^\star, s_{t+1}) \leq D_\psi(u^\star, s_t)$$

and

$$\sum_{t=1}^T D_\psi(s_{t+1}, s_t) \leq D_\psi(u^\star, s_1).$$

Proof. By Theorem D.2,

$$D_\psi(u^\star, s_{t+1}) + D_\psi(s_{t+1}, s_t) \leq D_\psi(u^\star, s_t).$$

Dropping the second term gives the first claim. Summing gives

$$\begin{aligned}
\sum_{t=1}^T D_\psi(s_{t+1}, s_t) &\leq \sum_{t=1}^T \left(D_\psi(u^\star, s_t) - D_\psi(u^\star, s_{t+1}) \right) \\
&= D_\psi(u^\star, s_1) - D_\psi(u^\star, s_{T+1}) \\
&\leq D_\psi(u^\star, s_1).
\end{aligned}$$

□

Theorem D.4 (Soft projection inequality)**.** *Let $\ell_t : S \to \mathbb{R}$ be convex and let*

$$s_{t+1} = \operatorname*{argmin}_{s \in S} \{D_\psi(s, s_t) + \eta_t \ell_t(s)\}.$$

Then for all $u \in S$,

$$\eta_t(\ell_t(s_{t+1}) - \ell_t(u)) \leq D_\psi(u, s_t) - D_\psi(u, s_{t+1}) - D_\psi(s_{t+1}, s_t).$$

Proof. For some $g_{t+1} \in \partial \ell_t(s_{t+1})$, optimality gives

$$\langle \nabla\psi(s_{t+1}) - \nabla\psi(s_t) + \eta_t g_{t+1}, u - s_{t+1} \rangle \geq 0.$$

Thus

$$\eta_t \langle g_{t+1}, s_{t+1} - u \rangle \leq \langle \nabla\psi(s_t) - \nabla\psi(s_{t+1}), s_{t+1} - u \rangle .$$

Convexity gives

$$\ell_t(s_{t+1}) - \ell_t(u) \leq \langle g_{t+1}, s_{t+1} - u \rangle .$$

Lemma D.1 gives

$$\langle \nabla\psi(s_t) - \nabla\psi(s_{t+1}), s_{t+1} - u \rangle = D_\psi(u, s_t) - D_\psi(u, s_{t+1}) - D_\psi(s_{t+1}, s_t).$$

Combining proves the claim. □

Theorem D.5 (KL consistency projection)**.** *Let p_t be strictly positive on a finite space $\mathcal{X}$. The solution of*

$$\operatorname*{argmin}_{p : \mathbb{E}_p[\phi] = c} D_{\mathrm{KL}}(p \| p_t)$$

has the form

$$p_{t+1}(x) = \frac{p_t(x) \exp(\lambda^\top \phi(x))}{Z(\lambda)}.$$

Proof. The Lagrangian is

$$\mathcal{L}(p, \lambda, \nu) = \sum_x p(x) \log \frac{p(x)}{p_t(x)} + \lambda^\top \left(\sum_x p(x)\phi(x) - c \right) + \nu \left(\sum_x p(x) - 1 \right).$$

Stationarity gives

$$\log \frac{p(x)}{p_t(x)} + 1 + \lambda^\top \phi(x) + \nu = 0.$$

Hence, after absorbing signs into λ and normalizing,

$$p_{t+1}(x) = \frac{p_t(x) \exp(\lambda^\top \phi(x))}{Z(\lambda)}.$$

□

Proposition D.6 (Gradient descent as linearized projection)**.** *For $D_\psi(s, s_t) = \|s - s_t\|_2^2/2$, replacing $\ell_t(s)$ in the soft projection objective by its first-order approximation at s_t gives $s_{t+1} = s_t - \eta\nabla\ell_t(s_t)$.*

Proof. The linearized objective is

$$\frac{1}{2}\|s - s_t\|^2 + \eta\langle\nabla\ell_t(s_t), s - s_t\rangle + \eta\ell_t(s_t).$$

The stationarity condition is

$$s - s_t + \eta\nabla\ell_t(s_t) = 0.$$

Thus

$$s_{t+1} = s_t - \eta\nabla\ell_t(s_t).$$

□

E Conservative Memory Proofs

Theorem E.1 (Exact associative binding, formal version of Theorem 3.2)**.** *Let $K \in \mathbb{R}^{r\times T}$ and $V \in \mathbb{R}^{m\times T}$. The set $\mathcal{C} = \{W \in \mathbb{R}^{m\times r} : WK = V\}$ is nonempty if and only if $\ker K \subseteq \ker V$. If $\mathcal{C}$ is nonempty, then*

$$\operatorname*{argmin}_{W\in\mathcal{C}} \frac{1}{2}\|W - W_0\|_F^2 = W_0 + (V - W_0K)K^+.$$

Proof. If $WK = V$ and $a \in \ker K$, then

$$Va = WKa = 0.$$

Thus $\ker K \subseteq \ker V$.

Assume $\ker K \subseteq \ker V$. Since K^+K is the orthogonal projector onto $(\ker K)^\perp$, the condition implies

$$V = VK^+K.$$

Thus $W = VK^+$ satisfies $WK = V$.

Let $R = V - W_0K$ and $W = W_0 + \Delta$. The projection problem is

$$\min_\Delta \frac{1}{2}\|\Delta\|_F^2 \quad \text{subject to} \quad \Delta K = R.$$

The minimum-norm solution is

$$\Delta^\star = RK^+.$$

Therefore

$$W^\star = W_0 + (V - W_0K)K^+.$$

□

Theorem E.2 (Innovation decomposition, formal version of Theorem 3.3)**.** *Assume $WK = V$. Let $\Pi = KK^+$, $z = (I - \Pi)k$, and $a = K^+k$. Then*

$$Wk = Va + Wz.$$

If $z = 0$, Wk is fixed over all old-feasible W. If $z \neq 0$, the constraint $Wk = v$ is equivalent over old-feasible W to $Wz = v - Va$.

Proof. The decomposition is

$$k = KK^+k + (I - KK^+)k = Ka + z.$$

Multiplying by W gives

$$Wk = WKa + Wz = Va + Wz.$$

If $z = 0$, then $Wk = Va$ for every W satisfying $WK = V$. If $z \neq 0$, then

$$Wk = v \quad \Longleftrightarrow \quad Va + Wz = v \quad \Longleftrightarrow \quad Wz = v - Va.$$

□

Theorem E.3 (Conflict impossibility, formal version of Theorem 3.4)**.** *Assume $W_tK = V$ and $z = (I - KK^+)k = 0$. If $W_t + \Delta$ preserves old evidence, then $(W_t + \Delta)k = W_tk$. Hence $Wk = v$ is compatible with old evidence if and only if $v = W_tk$.*

Proof. Preserving old evidence means

$$(W_t + \Delta)K = V = W_tK.$$

Thus

$$\Delta K = 0.$$

If $z = 0$, then $k \in \mathrm{span}(K)$, hence

$$\Delta k = 0.$$

Therefore

$$(W_t + \Delta)k = W_tk.$$

□

Theorem E.4 (Hard Conservative Projection Memory)**.** *Assume $W_tK = V$, let $\Pi = KK^+$, $z = (I - \Pi)k \neq 0$, and $e = v - W_tk$. Then*

$$W_{t+1} = W_t + \frac{ez^\top}{\|z\|^2}$$

preserves $WK = V$, satisfies $Wk = v$, and minimizes $\|W - W_t\|_F^2/2$ among all memories satisfying both constraints.

Proof. Let $\Delta^\star = ez^\top / \|z\|^2$. Since $z \perp \mathrm{span}(K)$,

$$\Delta^\star K = 0.$$

Thus

$$W_{t+1}K = W_t K = V.$$

Also $z^\top k = \|z\|^2$, so

$$\Delta^\star k = e.$$

Hence

$$W_{t+1}k = W_t k + e = v.$$

For minimality, any feasible update Δ satisfies

$$\Delta K = 0, \qquad \Delta k = e.$$

Since $\Delta K = 0$,

$$\Delta k = \Delta(I - \Pi)k = \Delta z.$$

Thus the problem reduces to

$$\min_{\Delta} \frac{1}{2} \|\Delta\|_F^2 \quad \text{subject to} \quad \Delta z = e.$$

The minimum-norm solution is $\Delta^\star = ez^\top / \|z\|^2$. □

Theorem E.5 (Soft Conservative Projection Memory, formal version of Theorem 3.5)**.** *Assume $W_t K = V$, let $\Pi = KK^+$, $z = (I - \Pi)k \neq 0$, and $e = v - W_t k$. The unique solution of*

$$\operatorname*{argmin}_{W:WK=V} \left\{ \frac{1}{2} \|W - W_t\|_F^2 + \frac{\eta}{2} \|Wk - v\|^2 \right\}$$

is

$$W_{t+1} = W_t + \frac{\eta}{1 + \eta \|z\|^2} ez^\top.$$

Moreover,

$$W_{t+1}K = V, \qquad v - W_{t+1}k = \frac{1}{1 + \eta \|z\|^2} e.$$

Proof. Write $W = W_t + \Delta$. The constraint $WK = V$ is equivalent to

$$\Delta K = 0.$$

Since $\Delta K = 0$,

$$\Delta k = \Delta(I - \Pi)k = \Delta z.$$

The objective becomes

$$\min_{\Delta:\Delta K=0} \frac{1}{2}\|\Delta\|_F^2 + \frac{\eta}{2}\|\Delta z - e\|^2 .$$

The stationarity equation in the feasible subspace is

$$\Delta + \eta(\Delta z - e)z^\top = 0.$$

Let $\Delta = \alpha e z^\top$. Substitution gives

$$\alpha e z^\top + \eta(\alpha e \|z\|^2 - e)z^\top = 0.$$

Thus

$$\alpha(1 + \eta\|z\|^2) = \eta.$$

Therefore

$$\Delta = \frac{\eta}{1+\eta\|z\|^2} e z^\top .$$

Since $z^\top K = 0$, $W_{t+1}K = V$. The residual is

$$\begin{aligned} v - W_{t+1}k &= e - \Delta k \\ &= e - \Delta z \\ &= e - \frac{\eta\|z\|^2}{1+\eta\|z\|^2} e \\ &= \frac{1}{1+\eta\|z\|^2} e. \end{aligned}$$

□

F Ridge and Bayesian Projection Memory Proofs

Theorem F.1 (Ridge solution)**.** *Let*

$$J_t(W) = \frac{\lambda}{2}\|W - W_0\|_F^2 + \frac{1}{2}\sum_{\tau=1}^{t} \rho_\tau \|W k_\tau - v_\tau\|^2 .$$

Define

$$A_t = \lambda I + \sum_{\tau=1}^{t} \rho_\tau k_\tau k_\tau^\top, \qquad B_t = \lambda W_0 + \sum_{\tau=1}^{t} \rho_\tau v_\tau k_\tau^\top .$$

Then the unique minimizer is $W_{t+1} = B_t A_t^{-1}$.

Proof. Differentiate:

$$\nabla J_t(W) = \lambda(W - W_0) + \sum_{\tau=1}^{t} \rho_\tau (W k_\tau - v_\tau) k_\tau^\top .$$

At optimum,

$$0 = \lambda(W - W_0) + \sum_{\tau=1}^{t} \rho_\tau (W k_\tau - v_\tau) k_\tau^\top.$$

Thus

$$W \left(\lambda I + \sum_{\tau=1}^{t} \rho_\tau k_\tau k_\tau^\top \right) = \lambda W_0 + \sum_{\tau=1}^{t} \rho_\tau v_\tau k_\tau^\top.$$

Since $\lambda > 0$, A_t is invertible. Hence

$$W_{t+1} = B_t A_t^{-1}.$$

□

Theorem F.2 (Recursive ridge update)**.** *Let $P_t = A_{t-1}^{-1}$ and $W_t = B_{t-1} P_t$. Define*

$$g_t = \frac{\rho_t P_t k_t}{1 + \rho_t k_t^\top P_t k_t}.$$

Then

$$W_{t+1} = W_t + (v_t - W_t k_t) g_t^\top, \qquad P_{t+1} = P_t - \frac{\rho_t P_t k_t k_t^\top P_t}{1 + \rho_t k_t^\top P_t k_t}.$$

Proof. The Sherman-Morrison identity gives

$$P_{t+1} = A_t^{-1} = P_t - \frac{\rho_t P_t k_t k_t^\top P_t}{1 + \rho_t k_t^\top P_t k_t}.$$

Let $a_t = k_t^\top P_t k_t$. Then

$$k_t^\top P_{t+1} = k_t^\top P_t - \frac{\rho_t a_t k_t^\top P_t}{1 + \rho_t a_t} = \frac{k_t^\top P_t}{1 + \rho_t a_t}.$$

Using $B_t = B_{t-1} + \rho_t v_t k_t^\top$,

$$\begin{aligned}
W_{t+1} &= B_t P_{t+1} \\
&= B_{t-1} P_{t+1} + \rho_t v_t k_t^\top P_{t+1} \\
&= W_t - \frac{\rho_t W_t k_t k_t^\top P_t}{1 + \rho_t a_t} + \frac{\rho_t v_t k_t^\top P_t}{1 + \rho_t a_t} \\
&= W_t + (v_t - W_t k_t) \frac{\rho_t k_t^\top P_t}{1 + \rho_t a_t} \\
&= W_t + (v_t - W_t k_t) g_t^\top.
\end{aligned}$$

□

Theorem F.3 (Bayesian ridge projection memory, formal version of Theorem 3.6)**.** *Assume each row w_j of W has prior $w_j \sim \mathcal{N}(w_{0,j}, \lambda^{-1} I)$ and observations satisfy $v_t = W k_t + \varepsilon_t$ with $\varepsilon_t \sim \mathcal{N}(0, \rho_t^{-1} I_m)$ independently over time. Then the posterior mean and MAP estimate are $B_t A_t^{-1}$.*

Proof. The negative log posterior, up to constants independent of W, is

$$\frac{\lambda}{2}\|W - W_0\|_F^2 + \frac{1}{2}\sum_{\tau=1}^{t}\rho_\tau \|Wk_\tau - v_\tau\|^2 .$$

This is $J_t(W)$. The posterior is Gaussian because the prior is Gaussian and the observation model is linear Gaussian. Hence the posterior mean equals the MAP. By Theorem F.1, this value is

$$B_t A_t^{-1}.$$

□